

Krishna Parvataneni

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Google Scholar: <https://scholar.google.com/citations?user=Wonn1kAAAAJ&hl=en>

Github: <https://github.com/kveni12>

Education

Massachusetts Institute of Technology (MIT)

Sep 2023 – June 2026

Bachelor of Science in Artificial Intelligence and Decision-Making

Cambridge, MA

Coursework: Computer Vision, Machine Learning, Computability Theory, Algorithms, Biochemistry

GPA 5.0/5.0

Experience

Intuitive Surgical

Jun 2026 – Present

Data Engineering Intern

Sunnyvale, CA

- Architected a retrieval-augmented figure generation pipeline that transforms procedural data and physician queries into automatically generated visual explanations and analytical reports.
- Developed Infrastructure-as-Code solutions using Terraform to manage AWS, Databricks, and Artifactory resources, enabling scalable, reproducible cloud deployments.
- Built analytics frameworks for measuring downstream product utility and adoption, delivering actionable performance insights to engineering stakeholders.
- Contributed to the development of scalable data platforms supporting AI-enabled healthcare applications and clinical decision-support workflows.

Massachusetts General Hospital

Dec 2023 – Present

Undergraduate Researcher, Precision in Medical Imaging Lab

Boston, MA

- Developed LLM-powered clinical decision support pipelines using Qwen, Llama, and Mixtral to automate longitudinal management of abscess drains, DVT, and VTE patients from unstructured radiology reports, achieving physician-level extraction performance across multiple clinical variables.
- Built scalable data infrastructure for processing ~2,000 interventional radiology reports and generating structured patient tracking databases through parallelized inference workflows.
- Designed machine learning risk prediction models for post-procedural outcomes in hemodialysis patients, leveraging multicenter ACS-NSQIP data to identify factors associated with 30-day rehospitalization and complications.
- Conducted model evaluation, feature selection, explainability analyses (SHAP), and topological data analysis to characterize patient risk subgroups and improve clinical interpretability.

MIT Department of Mechanical Engineering

Jul 2023 – Present

Undergraduate Researcher, Traverso Lab

Cambridge, MA

- Designed and prototyped ingestible medical devices using SOLIDWORKS and Fusion 360, developing deployable gastrointestinal drug delivery systems for treatment of hyperoxaluria.
- Developed and experimentally validated a swallowable oxalate oxidase delivery platform, designing ELISA-based assays to quantify enzyme activity and therapeutic performance under simulated gastrointestinal conditions.
- Built a geometric deep learning framework for bio-inspired microneedle design generation, leveraging adaptive meshing, surface parameterization, graph neural networks, and geometric feature extraction to learn structure–function relationships from 3D geometries.

MIT McGovern Institute

Jul 2023 – Jan 2026

Undergraduate Researcher, Graybiel Lab

Cambridge, MA

- Analyzed large-scale neuronal recording datasets and behavioral kinematic trajectories to investigate neural mechanisms underlying decision-making in freely moving mice.
- Developed machine learning and statistical analysis pipelines for neural signal processing, behavioral state characterization, and multimodal data integration.
- Performed histological and protein expression analyses following viral injections into the striatum and substantia nigra to validate experimental targeting and circuit manipulation.

MIT-Harvard Health Sciences and Technology

Jul 2023 – Feb 2025

Undergraduate Researcher, Edelman Lab

Cambridge, MA

- Developed deep learning models using GANs and computer vision techniques to analyze echocardiography data for cardiac motion characterization and mechanical pacemaker research.
- Built image preprocessing, augmentation, and classification pipelines for ultrasound-based cardiac phenotyping.

Harvard T.H. Chan School of Public Health

Jul 2023 – Aug 2024

Teaching Assistant

Boston, MA

- Supported instruction of graduate-level machine learning and data science coursework in R for physicians and clinical researchers (BST.209).
- Collaborated with a multidisciplinary team of physicians to develop predictive models for emergency department disposition in congestive heart failure patients, emphasizing fairness-aware modeling and bias mitigation strategies.
- Designed educational materials and coding exercises covering machine learning workflows, model evaluation, and clinical applications of data science

Kellis Lab, MIT Computer Science and AI Laboratory

Aug 2023 – Feb 2024

Research Intern

Cambridge, MA

- Developed single-cell sequencing analysis pipelines to identify structural genomic variants associated with neuropsychiatric disease.

Extracurriculars**Harvard Global Alliance for Medical Innovation**

Aug 2024 – Present

Co-President, Project Team Lead

Cambridge, MA

- Led an interdisciplinary organization supporting AI and healthcare innovation projects across medical imaging, digital health, and global health applications.
- Managed development of a deep learning-based fracture detection platform for pelvic and acetabular X-rays, designed for deployment in resource-constrained healthcare settings in Malawi.
- Mentored teams building machine learning systems for cardiovascular imaging analysis, clinical decision support, and healthcare accessibility initiatives.
- Established collaborations with physicians, researchers, and healthcare organizations to guide model development, validation, and real-world implementation.
- Oversaw project roadmaps, technical reviews, and team coordination across multiple concurrent research and product development efforts.

Skills and Interests

Technical: Python, Terraform, TensorFlow, PyTorch, Keras, Large Language Models (LLAMA, GPT, Mixtral, Qwen), vLLM, R, Java, C, C++, Q#, SQL, Flutter, JavaScript, HTML/CSS, SolidWorks, FUSION360

Laboratory: immunohistochemical staining, ELISA, PCR, brain sectioning, animal handling (pigs/mice)

Language: Telugu (bilingual proficiency), Spanish (working proficiency), Italian (elementary proficiency)